Exotic random variables

Gareth Tribello[1]

[1]School of Mathematics and Physics

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Abstract

Samples of multiple 1D and 2D random vectors were generated. The distributions of the distances and angles between pairs of these random vectors were then investigated. In the final part of the report we then extended our investigation and studied the distribution of the area and internal angles of random triangles that were formed from three random 2D vectors.

1 Introduction

In other exercises and example reports we have generated samples of random variables and performed analysis upon them to compute means and histograms. In this example report we will explore other forms of analysis that we can perform on the samples of random variables. As we will see, these analysis all work by taking pairs or triples of the random variables from our sample and by computing new random variables from these sets of random variables.

All these analysis will be performed on samples from the two distributions shown in figure 1. The random variables that were sampled to construct these two estimates of the histogram shown in this figure are:

1. A uniform (continuous) random variable with $a = -1$ and $b = 1$.
2. A normal random variable with $\mu = 0$ and $\sigma = 1/3$.

The normal distribution has infinite support, while the uniform distribution is only non-zero for values between -1 and +1. As you can see from the distributions in figure 1 setting $\sigma = 1/3$ ensures that the majority of the normal random variables that we generate will fall between -1 and +1. This ensures that the random variables that we generate in later parts of the report can be compared on the same scales.

2 Pairs of random variables in 1D

If we have a sample of n identical random variables in the random column vector $\mathbf{X}$ we can generate a symmetric matrix of random variables $\mathbf{Y}$ using:

$$\mathbf{Y} = \mathbf{X} \ominus \mathbf{X}^T$$

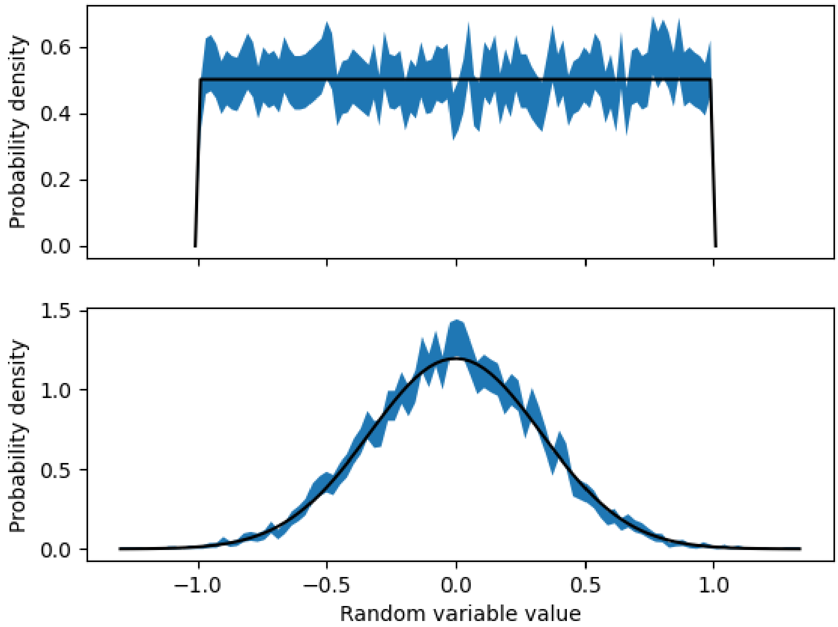

Fig. 1. The two basic distributions that have been used to generate all the random variables that have been studied in this work. The blue shaded area in the top panel indicates the 90% confidence limit on an estimate of the histogram for a uniform (continuous) random variable with $a = -1$ and $b = 1$ that was obtained by sampling the random variable 10000 times. The blue shaded area in the second figure shows an estimate of the histogram for a normal random variable with $\mu = 0$ and $\sigma = 1/3$ that was obtained by a similar strategy. The black lines are the true probability density functions for these random variables.

where the $\ominus$ sign here indicates that we are computing element Y_{ij} from X_i and X_j using:

$$Y_{ij} = X_i - X_j.$$

The figures in the left column of 2 shows the distribution of the Y_{ij} random variables that are generated by this procedure when the variables in the vector $\mathbf{X}$ are uniform random variables (top) and normal random variables (bottom). To generate these figures we generated $n = 600$ random variables when constructing the vector $\mathbf{X}$. The histograms were then constructed from the $n(n-1)/2$ random variables in the upper triangle of the matrix $\mathbf{Y}$.

The left column of figure 2 shows that if the vector $\mathbf{X}$ is filled with uniform random variables then the distribution of the random variables in the matrix $\mathbf{Y}$ does not resemble that of the random variables in $\mathbf{X}$. However, if $\mathbf{X}$ is full of normal random variables the distribution for the random variables in $\mathbf{Y}$ resembles the bell shaped curve that each of the variables in $\mathbf{X}$ is a sample from.

These results that we obtain from sampling are consistent with results for the probability density of this type of random variable that can be derived analytically. For example, we know that if $Y = a_1X_1 + a_2X_2$ where $X_1 \sim N(\mu_1, \sigma_1)$ and $X_2 \sim N(\mu_2, \sigma_2)$ and X_1 and X_2 are

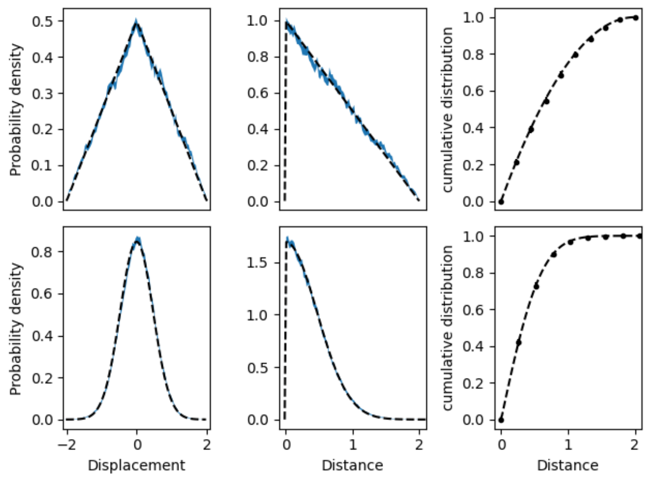

Fig. 2. Distributions of displacement (left column) and distance (middle column) between pairs of uniform (top row) and normal (bottom row) random variables. The blue shaded regions show the 90% confidence limit on an estimates of the probability density that were obtained by sampling $n = 600$ of the underlying random variables and computing the $n(n-1)/2$ instances of the distance/displacement from these random variables. The points and error bars in the figures in the right most column show estimates of the cumulative distribution that were obtained using the method described in the main text. The error bars on these estimates are often smaller than the points. The dashed lines in the panels are plots of the analytic expressions for the probability densities and cumulative distributions for these types of random variable that are discussed in the main text.

independent then $Y \sim N\left(a_1\mu_1 + a_2\mu_2, \sqrt{a_1^2\sigma_1^2 + a_2^2\sigma_2^2}\right)$. If the random variables in $\mathbf{X}$ are samples from a normal distribution with $\mu = 0$ and $\sigma = 1/3$ then the random variables in $\mathbf{Y}$ are samples from a normal distribution with $\mu = 0$ and $\sigma = \sqrt{2(1/3)^2}$. Meanwhile, if the random variables in $\mathbf{X}$ are uniform (continuous) random variables then the random variables in $\mathbf{Y}$ are samples from the probability density:

$$f(y) = \begin{cases} \frac{1-|y|/2}{2} & \text{for} \quad |y| < 2 \\ 0 & \text{otherwise} \end{cases}$$

These distributions are indicated with black dashed lines in figure 2. You can see that there is close agreement with the distributions that we obtain from sampling and these analytic results.

The middle column of figure 2 shows what is observed when we compute the matrices of random variables using:

$$\mathbf{D} = \left|\mathbf{X} \ominus \mathbf{X}^T\right|$$

In other words, to construct the estimates of the histograms in the middle panel of figure 2 we calculate the matrix of distances (the modulus of the displacement) between the pairs random variables. Obviously, if the probability density function for the random variables in $\mathbf{Y}$ is $f(y)$, the probability density for the random variables in $\mathbf{D}$ is:

$$p(z) = \begin{cases} f(-z) + f(z) & \text{if} \quad z > 0 \\ 0 & \text{otherwise} \end{cases}$$

These analytic distributions are shown with the black dashed lines in figure 2 and you can see that once again there is a close match between the analytic results and the results that we obtain from sampling.

Notice, that we can also estimate the cumulative distribution, $F(r)$, for the random variable Z by computing the following quantity from our sample of n random variables:

$$F(r) = \frac{1}{n}\sum_{i=1}^{n} B_i \tag{1}$$

where the B_i in this expression are the elements of the n dimensional vector B that is computed as follows:

$$\mathbf{B} = \mathbf{C}\mathbf{1} \qquad \text{where} \qquad C_{ij} = \begin{cases} 1 & \text{if} \quad D_{ij} \le r \\ 0 & \text{otherwise} \end{cases} \tag{2}$$

In this expression $\mathbf{1}$ is an n-dimensional vector in which every element is equal to one and the D_{ij} represent the elements of the $n \times n$ matrix of distances that was defined above. The points in the right column of figure 2 are the estimates that we obtained for the cumulative probability distributions that we obtained using this procedure. You can see that these estimates are in close agreement with the analytic result for the distribution that is being sampled.

3 Pairs of random variables in 2D

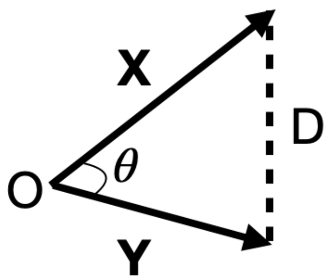

Fig. 3. Diagram illustrating how the cosine rule emerges when we calculate the distance between the end points of two vectors $\mathbf{X}$ and $\mathbf{Y}$.

Suppose that $\mathbf{X}$ and $\mathbf{Y}$ are the two vectors shows in figure 3. Each of the vectors is a pair of random variables. We can calculate the square of the distance between these two points in the plane using:

$$D^2 = \sum_{i=1}^{2}(X_i - Y_i)^2 = \sum_{i=1}^{2} X_i^2 + Y_i^2 - 2X_iY_i$$

The result after the second equals sign here can be written in matrix form as follows:

$$D^2 = \mathbf{X}^T\mathbf{X} + \mathbf{Y}^T\mathbf{Y} - 2\mathbf{X}^T\mathbf{Y} \tag{3}$$

Furthermore, this equation holds regardless even if the vectors $\mathbf{X}$ and $\mathbf{Y}$ have $n > 2$ elements.

To understand this equation remember that if $\mathbf{X}$ and $\mathbf{Y}$ are column vectors then $\mathbf{X}^T\mathbf{Y}$ is their dot product. The dot product of the vector $\mathbf{X}$ with itself, $\mathbf{X}^T\mathbf{X}$, is the square of the modulus, $|\mathbf{X}|$, of the vector. Furthermore, the dot product $\mathbf{X}^T\mathbf{Y}$ can also be calculated as $|\mathbf{X}||\mathbf{Y}|\cos(\theta)$, where $|\mathbf{X}|$ is the modulus of the vector $\mathbf{X}$ and θ is the angle between the vectors $\mathbf{X}$ and $\mathbf{Y}$. Substituting these realizations into equation 3 brings us to:

$$D^2 = |\mathbf{X}|^2 + |\mathbf{Y}|^2 - 2|\mathbf{X}||\mathbf{Y}|\cos(\theta)$$

which is an old friend from GCSE trigonometry - the cosine rule.

Now suppose that $\mathbf{X}$ is an $m \times n$ matrix and that each of its n columns contains an m-dimensional vector of random numbers. We can use the logic of the derivation above to compute the matrix, $\mathbf{D}$, of distances between these random points as:

$$\mathbf{D} = \sqrt{\mathbf{m} \oplus \mathbf{m}^T - 2\mathbf{X}^T\mathbf{X}} \qquad \text{where} \qquad \mathbf{m} = (\mathbf{X} \odot \mathbf{X})^T \mathbf{1}$$

In this expression, $(\mathbf{X} \odot \mathbf{X})_{ij}$ is equal to X_{ij}^2 and $\mathbf{1}$ is an m-dimensional column vector in which every element is equal to one so $\mathbf{m}$ is an n-dimensional vector whose ith element is equal to the square modulus of the vector in the ith column of $\mathbf{X}$. We then use the $\oplus$ symbol much as we used the $\ominus$ symbol in the last section so $\mathbf{m} \oplus \mathbf{m}^T$ is an $n \times n$ matrix with:

$$\mathbf{m} \oplus \mathbf{m}^T = \sum_{k=1}^{m} X_{ki}^2 + \sum_{k=1}^{m} X_{kj}^2 \tag{4}$$

Lastly, note that $\mathbf{X}^T\mathbf{X}$ is also an $n \times n$ matrix and that:

$$\left(\mathbf{X}^T\mathbf{X}\right)_{ij} = \sum_{k=1}^{m} X_{ik}X_{kj}$$

which is the dot product of the vectors in columns i and j of $\mathbf{X}$.

To construct figure 4 we used this logic to analyze the distribution of distances between pairs of normal and uniform continuous random variables that were sampled from the distributions described in the introduction. In other words, we generated samples of $n = 1000$ pairs of normal and uniform random variables which we put in a 2×1000-dimensional matrix. We then

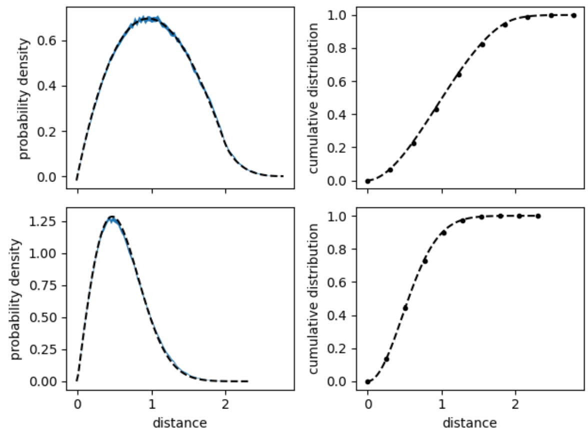

Fig. 4. Probability density functions for distribution of distances (left column) between pairs of 2-dimensional vectors of uniform continuous (top) and normal (bottom) random variables. The blue shaded histograms indicate the 90% confidence interval on the estimate of the distributions were obtained when we constructed a histogram from $n(n-1)/2$ distances that can be extracted from a sample of $n = 1000$ pairs of random variables. The black points in the figures in right column are estimates of the cumulative probability distribution that were obtained using the procedure described in the text. Confidence limits on these estimates of the probability were computed by these are smaller that the points used to represent the estimates. Black dashed lines are plots of the analytic expressions for these probability density functions and cumulative probabilities.

used equation 4 to calculate the 1000×1000 matrix containing the distances between each pair of points. The left column of figure 4 shows the estimates of the probability distribution for this random variables that we obtained from the $n(n-1)/2$ samples in the upper triangle of this matrix.

Analytic expressions for the probability distribution the the distance between the end points of two 2-dimensional random vectors are known. If the random variables are uniform (continuous) random variables with $a = -1$ and $b = 1$ then the probability density is:

$$f(x) = \begin{cases} \frac{x}{2}\left(\pi - 2x + \frac{x^2}{4}\right) & \text{if} \quad 0 < x < 2 \\ \frac{x}{2}\left[4\arcsin\left(\frac{2}{x}\right) + 2\sqrt{x^2-4} - \pi - 2 - x\right] & \text{if} \quad 2 \leq x \leq \sqrt{8} \\ 0 & \text{otherwise} \end{cases}$$

If we compute $\sqrt{\sum_{i=1}^m A_i}$ from a sample of m normal random variables, A_i, with $\mu = 0$ and $\sigma = 1$ we get a sample from a chi-distribution with m degrees of freedom. However, the random variable we are sampling here is given by:

$$D = \sqrt{(X_1 - X_2)^2 + (X_3 - X_4)^2}$$

where each X_i is a normal random variable with $\mu = 0$ and $\sigma = 1/3$. As we learned in the last section, $(X_1 - X_2)$ and $(X_3 - X_4)$ are samples from a normal distribution with $\mu = 0$

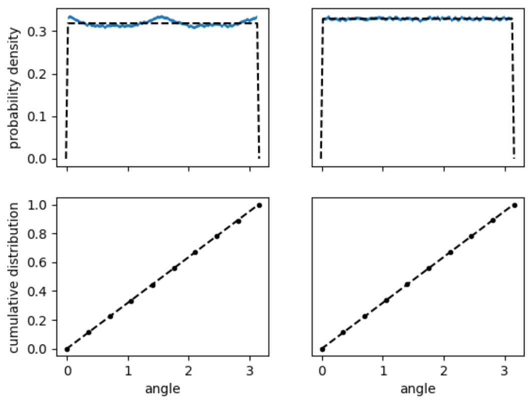

Fig. 5. Probability density functions (top row) for distributions of angles between pairs of 2-dimensional vectors of uniform continuous (left column) and normal (right column) random variables. The blue shaded histograms were obtained when we constructed a histogram from the $n(n-1)/2$ angles that can be extracted from a sample of $n = 2000$ random variables. When these are compared with the probability density function for a uniform (continuous) random variable with $a = 0$ and $b = \pi$ (black dashed line you can see that there are certain preferred angles for the distribution that was obtained from the uniform random variables. However, for angles generated from normal random variables the distribution is uniform. Interestingly, when you try to estimate the cumulative probability distribution for the angle using a variant on the technique introduced in equations 1 and 2 (black points in bottom panels) you do not see the difference between the distribution obtained by sampling and the analytic expressions for the cumulative probability distribution for a uniform random variable with $a = 0$ and $b = \pi$ (black dashed lines). 90% confidence limits on the estimates of the cumulative distribution in these bottom panels were calculated. They are smaller than the symbols.

and $\sigma = \sqrt{2/9}$) so we have to scale the chi distribution with 2 degrees of freedom by $\sqrt{2/9}$ to arrive at the probability density function for D. These functions are plotted using dashed black lines in figure 4.

We can also use the procedure that was outlined in equations 1 and 2 in the last section to calculate estimates for the cumulative distributions of these functions from the matrix of distances between our random vectors $\mathbf{D}$. The results from this analysis are shown using black points for the panels in the right column of figure 4. You can see that there is good agreement for the estimates of the cumulative distribution that we get by applying this procedure and the analytic results for the cumulative distributions that we get by integrating the probability densities.

Instead of calculating the distances between our vectors we can compute the angles θ (shown in figure 3) between them. To compute the $n \times n$ matrix, $\mathbf{\Theta}$, of these angles from the $m \times n$ matrix, $\mathbf{X}$, of vectors you do the following:

$$\mathbf{\Theta} = \arccos\left(\frac{\mathbf{X}^T\mathbf{X}}{\sqrt{\mathbf{m} \otimes \mathbf{m}^T}}\right). \tag{5}$$

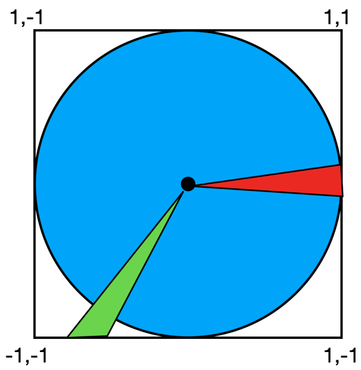

Fig. 6. The reason why there are preferential directions when you generate random vectors using uniform (continuous) random variables. As the distribution is uniform, the green and red shaded areas are proportional to the probability of generating an angle that is within a certain range of values. You can clearly see that the likelihood of generating a vector that points towards one of the corners of the square is greater than generating a vector that points to the center of the sides of the square. The directions would be uniformly sampled if you rejected any pairs of random variables that did not fall within the blue circle.

The elements of the $n \times n$ matrix in the square root of the denominator of this expression are given by:

$$\mathbf{m} \otimes \mathbf{m}^T = \left(\sum_{k=1}^{m} X_{ki}^2 \right) \left(\sum_{k=1}^{m} X_{kj}^2 \right)$$

and the numerator is the $n \times n$ matrix of dot products between vectors. Square root, arc cosine and division operations are applied to the matrices element wise.

Samples of $n = 2000$ pairs of normal and uniform random variables were generated from the distributions described in the introduction and placed in a two 2×2000 $\mathbf{X}$ matrices. The $n(n-1)/2$ random angles in the upper triangle of the symmetric matrix $\mathbf{\Theta}$ were then used to construct the estimates of the histograms shown in the top row of figure 5. You can see that the all the sampled angles fall between 0 and π as we would expect. Furthermore, for the normal distribution the angle is uniformly distributed across this range as one would expect. However, when uniform (continuous) random variables are generated we see deviations from a uniform distribution because, as figure 6, shows there is a greater likelihood of generating random variables that point towards the corners of the domain of the probability density function for this random variable.

Lastly, note that even though $\mathbf{\Theta}$ is a matrix of angles and not distances we can still use the trick introduced in equations 1 and 2 to estimate the cumulative distribution function for the angles. The black points in the lower panel of figure 5 show estimates for the cumulative distribution for the distributions of angles when our samples are uniform (left) and normal (right) random variables. When this analysis is performed the differences between distribution of angles that is generated from the uniformly generated random vectors and the cumulative distribution for a uniform (continuous) random variable with $a = 0$ and $b = \pi$ are not clear.

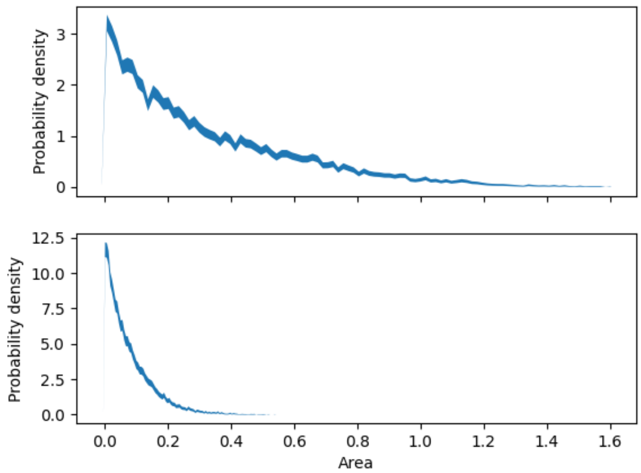

Fig. 7. Estimates of the probability distributions for the area of a randomly generated triangle. To construct each histogram we generated $n = 50$ pairs of uniform continuous (top panel) and normal (bottom panel) random variables. The histogram was then generated using the areas of the $n(n-1)(n-2)/6$ distinct triangles that can be constructed from these n random vectors. The blue shaded area indicates the 90% confidence limit on our estimate of the probability density function.

4 Triplets of random variables in 2D

Suppose that we generate n pairs of random variables, $\{X, Y\}_i$, and place each of these pairs in a set, S, with n elements. We can construct the bounded power set $\mathcal{P}_3(S)$ from S, which will contain all the $n(n-1)(n-2)/6$ sets of vectors with cardinality 3. We can think of each element of $\mathcal{P}_3(S)$ as a 2×3 matrix, $\mathbf{M}^{(j)}$ that contains the vertices of a triangle in its three columns. We can calculate the area of this triangle using:

$$\mathbf{A}^{(j)} = \frac{1}{2}\left|\det\begin{pmatrix} M_{12}^{(j)} - M_{11}^{(j)} & M_{22}^{(j)} - M_{21}^{(j)} \\ M_{13}^{(j)} - M_{11}^{(j)} & M_{23}^{(j)} - M_{21}^{(j)} \end{pmatrix}\right|$$

Consequently, if we have n pairs of random variables we can calculate $n(n-1)(n-2)/6$ samples of this random variable.

Figure 7 shows estimates of the histogram for this random variable that we obtain by applying this procedure to a set of $n = 50$ pairs of uniform continuous (top) and normal (bottom) random variables. The histograms were thus constructed from $n(n-1)(n-2)/6$ samples of the area.

Figure 7 shows that small triangles occur more frequently than large triangles regardless whether the random vectors are generated from normal or uniform (continuous) random variables. Furthermore, the triangles generated using the normal random variables have areas that are (on average) smaller than those that are generated from the uniform (continuous)

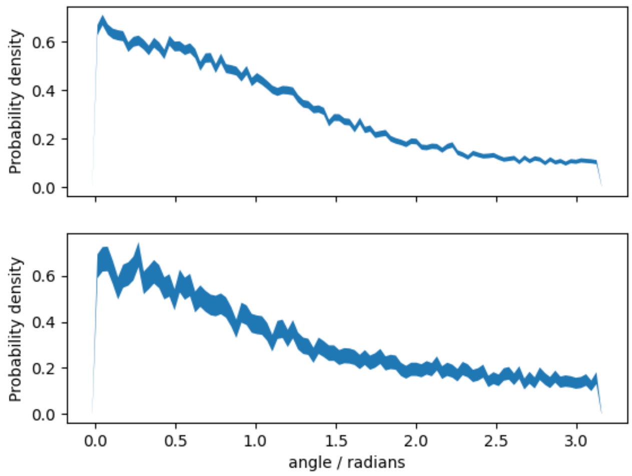

Fig. 8. Estimates of the probability distribution for an internal angle in a randomly generated triangle. To construct each histogram we generated $n = 50$ pairs of uniform (top panel) and normal (bottom panel) random variables. The histogram was then generated using the 3 internal angles of each of the $n(n-1)(n-2)/6$ random triangles that can be constructed from these n random vectors ($n(n-1)(n-2)/2$ samples in total). The blue shaded area indicates the 90% confidence limit on our estimate of the probability density function.

random variables. This makes sense as figure 4 shows that random points generated from normal random variables are more likely to be close together.

Instead of calculating the areas of the triangles formed by three random points we can calculate the 3 angles at the corners of our random triangles. If we have generated n random vectors we can generate $n(n-1)(n-2)/3$ samples of this quantity. Figure 8 shows the distributions of the angles that we obtained from the data that was used to construct figure 7. You can see that the distribution of angles that we obtain here is not even close to the uniform distribution that we obtained when we constructed figure 5.